

Physical Layer Fundamentals

Dr. Gunjan Chugh
Assistant Professor
DTU

Introduction: Data Communication System

Source is where data originates. It can be a computer, telephone handset, video camera, etc.

Figure: Simplified model of a data communication system



Transmitter

Converts data into signal using a device such as modem.

Communication Medium

Twisted-pair, coaxial cable, optical fiber or wireless.

Receiver

Converts signal back to data for destination.

Data and Signal

Data refers to information that conveys meaning. It comes in forms: text, graphics, audio, video.

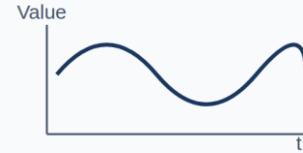
Signal is the electrical, electronic or optical representation of data.

Data Types:

Analog - continuous values

Digital - discrete values

Figure: Analog signal and Digital signal



Analog Signal



Digital Signal

Key Differences

Analog - continuous waveform varying smoothly

Digital - limited values, usually 0 and 1

Bandwidth and Channel Capacity

Bandwidth is the range of frequencies over which most of the signal energy is contained.

$$\text{Bandwidth (B)} = F_h - F_l$$

Digital signal requires a **low-pass channel** (0 to f). **Analog signal** requires a **band-pass channel** (f_1 to f_2).

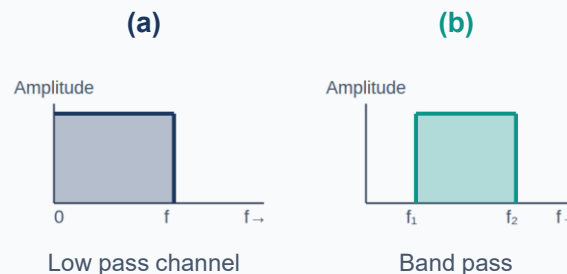


Figure: (a) Low-pass channel for digital signals (b) Band-pass channel for analog signals

Bandwidth of medium decides signal quality at the other end.

Bandwidth

Bandwidth is the **range of frequencies** over which a communication channel can transmit signals effectively.

$$B = f_{high} - f_{low}$$

Unit: Hertz (Hz)

Types of Bandwidth

(a) Analog Bandwidth

- Frequency range of a signal or channel
- Used in analog communication

Example:

Telephone channel

$$3400 - 300 = 3100 \text{ Hz}$$

(b) Digital Bandwidth

- Number of bits transmitted per second
- Measured in **bps**

Example:

Internet speed = **100 Mbps**

Why Bandwidth is Important?

- Determines **maximum data rate**
- Affects **signal quality**
- Higher bandwidth → more data transfer

Channel Capacity

Channel Capacity is the **maximum data rate** that can be transmitted **reliably** over a channel.

It depends on:

- Bandwidth
- Noise
- Signal power

Low-Pass Channel

- Allows **frequencies from 0 Hz (DC) up to some maximum frequency f**
- Blocks higher frequencies

Low-pass *channel*: $0 \rightarrow f$

Band-Pass Channel

- Allows **only a specific band of frequencies**
- Blocks frequencies **below f_1 and above f_2**

Band-pass *channel*: $f_1 \rightarrow f_2$

Why Digital Signals Need a Low-Pass Channel

- Digital signal = **0 and 1**
- Represented as **voltage levels**

Example:

- Logic 1 = +5V, Logic 0 = 0V

These voltage levels **do not oscillate around a carrier frequency.**

Key Reason

Since digital signals:

- Start from **0 Hz**
- Contain low-frequency components

Practical Example:

Ethernet cable, USB data, Computer LAN

All transmit **digital pulses directly**, without modulation.

Why Analog Signals Need a Band-Pass Channel ?

Analog signals are **continuous waveforms** , usually modulated on a **carrier frequency**

Example:

- Radio signal at **100 MHz**, TV signal at **500 MHz**

These signals **do not start from 0 Hz**.

Key Reason

Analog transmission:

- Uses **modulation**
- Shifts signal to **higher frequency range**

Hence, analog signals occupy a **band of frequencies (f_1 to f_2)** and need a **band-pass channel**.

Practical Example

- Radio broadcasting, Cable TV, Telephone voice transmission

Digital Signal: Bit Rate and Bit Interval

Most digital signals are **aperiodic**, so period and frequency are not appropriate. Two new terms are used:

BIT INTERVAL

Time required to send one single bit.

BIT RATE

Number of bits sent in one second (bps).

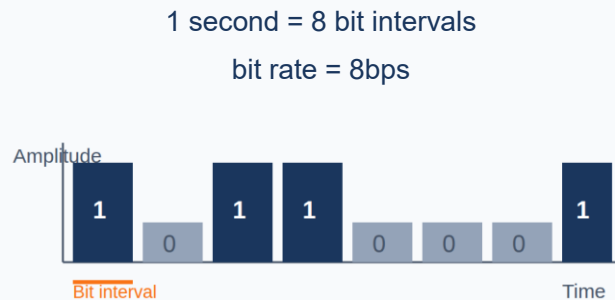


Figure: Bit Rate and Bit Interval

Relationship:

If bit interval = T , then Bit Rate = $1/T$ bps

Bit Rate

- **Bit Rate** is the **number of bits transmitted per second** over a communication channel.
- It tells **how fast data is sent**.

Unit:

- bits per second (**bps**)
- kbps, Mbps, Gbps

Formula

$$\text{Bit Rate} = \frac{\text{Number of bits}}{\text{Time (seconds)}}$$

Example 1

If 8000 bits are transmitted in 2 seconds:

$$\text{Bit Rate} = \frac{8000}{2} = 4000 \text{ bps}$$

Example 2 (Real life)

- Internet speed = **100 Mbps**
- Means **100 million bits** are transmitted every second

Bit Interval

- **Bit Interval** is the **time required to transmit one single bit**.
- It tells **how long one bit stays on the channel**.

Unit: seconds per bit (s/bit)

Formula

$$\text{Bit Interval} = \frac{1}{\text{Bit Rate}}$$

Relationship Between Bit Rate and Bit Interval

$$\text{Bit Rate} = \frac{1}{\text{Bit Interval}}$$

They are **inverse of each other**.

Example 3

Bit rate = **1000 bps**

$$\text{Bit Interval} = \frac{1}{1000} = 0.001 \text{ sec} = 1 \text{ ms}$$

Example 4

Bit interval = **0.0005 sec**

$$\text{Bit Rate} = \frac{1}{0.0005} = 2000 \text{ bps}$$

Note:

- **High Bit Rate** → bits sent **faster** → **small bit interval**
- **Low Bit Rate** → bits sent **slowly** → **large bit interval**

Advantages of Digital Transmission

Digital transmission has several advantages over analog transmission:

1. NOISE IMMUNITY

Digital amplifiers regenerate exact signal, eliminating cumulative errors.

2. VERSATILITY

Voice, data, video can all be carried by digital circuits using "1" and "0".

3. MULTIPLEXING

Easier to multiplex large channel capacities with digital

4. ENCRYPTION

Easy to apply encryption to digital data for security.

5. INTEGRATION

Better integration when all signals in one form.

Key Point: In digital circuits, incoming signal is sampled, value determined, and new signal generated.

Base-band and Broadband Signals

BASE-BAND

A baseband signal is a digital signal that is transmitted directly over the communication medium without using any carrier frequency.

- **Entire frequency spectrum** utilized.
- Transmission is **bi-directional**.

Base band is simple to install, interfaces inexpensive, but limited range.

BROADBAND

Uses **analog signaling** with carrier.

Uses **frequency division multiplexing**.

Signal propagates in **one direction** only.

Broadband is more complicated, expensive, but offers more services (audio, video).

Characteristics of Baseband Signals

- Digital in nature (0 and 1)
- Uses **low-pass channel** (0 to f)
- **Entire bandwidth** occupied by one signal
- Transmission is **bi-directional**
- Short distance communication

Example

- Ethernet (LAN)
- Computer to computer communication
- USB communication

Advantages

- Simple and inexpensive.
- Easy to install.
- High data rate for short distances.

Limitations

Limited range
Only one signal at a time
Not suitable for long-distance transmission.

Characteristics of Broadband Signals

- Analog in nature
- Uses **band-pass channel** (f_1 to f_2)
- Bandwidth divided into multiple channels
- Transmission is **one-directional**
- Suitable for long distances

Example

- Cable TV
- Radio broadcasting
- Internet over cable

Advantages

- Supports multiple signals simultaneously.
- Suitable for audio and video.
- Long-distance communication.

Limitations

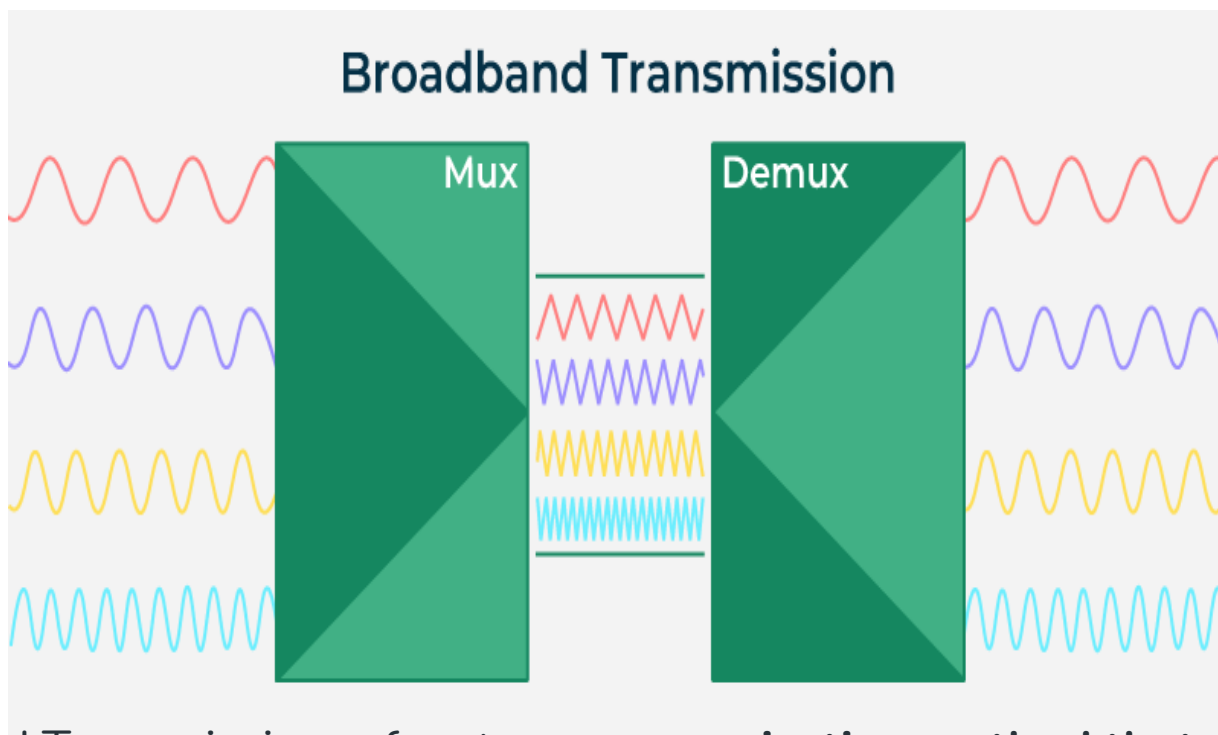
Expensive
Complex equipment
Difficult to manage and
troubleshoot

Baseband Transmission



Baseband Transmission refers to a method of sending digital signals over a communication medium using a single, low-frequency channel.

- In this type of transmission, the entire bandwidth of the medium is used for a single signal.



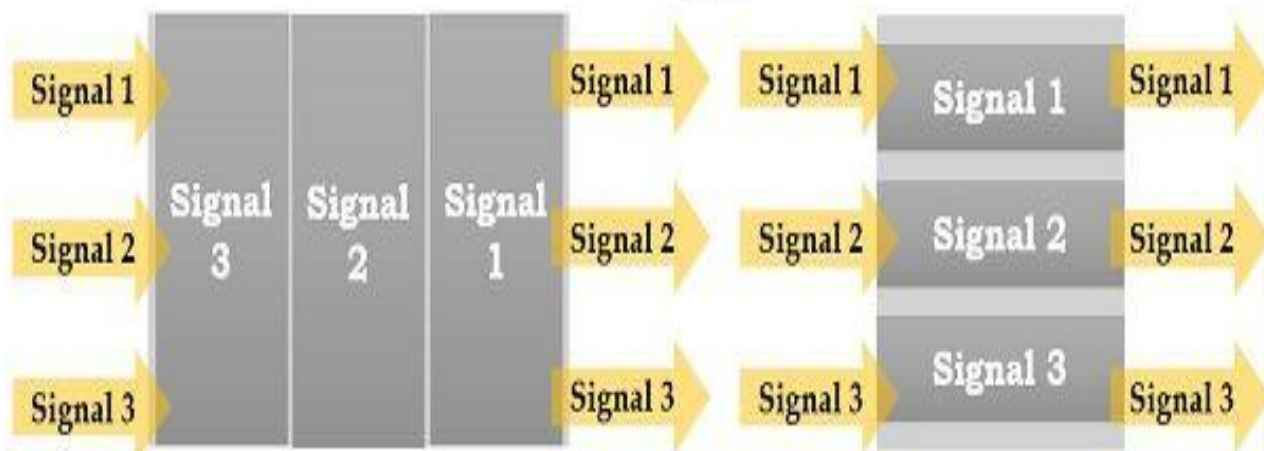
Broadband Transmission refers to a **communication method that uses a wide range of frequencies to transmit multiple signals simultaneously over a single medium.**

- Each frequency band carries a different signal, such as voice, video, or data. This allows high-speed data transmission and maximizes the use of available bandwidth.

Baseband Transmission

Vs

Broadband Transmission



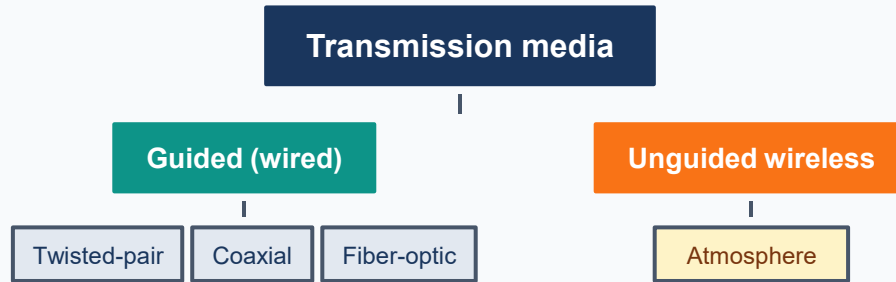
Baseband vs Broadband

Feature	Baseband	Broadband
Signal type	Digital	Analog
Channel type	Low-pass	Band-pass
Bandwidth usage	Entire bandwidth	Divided into bands
Multiplexing	TDM	FDM
Direction	Bi-directional	One-directional
Distance	Short	Long
Example	Ethernet LAN	Cable TV

Transmission Media

Transmission media is the physical path between transmitter and receiver.

Figure: Classification of the transmission media



GUIDED

Capacity depends on medium, length, point-to-point or multipoint.

UNGUIDED

Transmits electromagnetic signals but does not guide them.

Guided Media: Twisted Pair Cable

Two copper wires strung between two points. Wires are **"twisted" together in a helix** to reduce interference and cross-talk.

Characteristics:

Can carry both analog and digital signals

Data rates of several Mbps common

Spans distances of several kilometers

Good, low-cost communication

Max 1 km for data rates up to 1 Mb/s. Data rate inversely proportional to square of line length.



Figure: CAT5 cable (twisted cable)

UTP

Unshielded Twisted Pair
Telephone lines, LAN (100m)

STP

Shielded Twisted Pair
Metallic braid reduces
interference

Guided Media: Coaxial Cable

With "coax", medium consists of a **copper core** surrounded by insulating material and **braided outer conductor**.

BASE BAND COAXIAL

Digital transmission, 1 KHz to 20 MHz over 1 km.
10 Mbps typical.

BROADBAND COAXIAL

Analog, bandwidth 300 MHz, ~150 Mbps. Up to 100 km.

Uses: Cable TV (CATV), LAN distribution

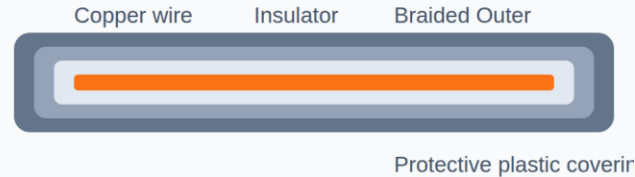


Figure: Co-axial cable

Characteristics

Superior frequency characteristics compared to twisted-pair. Used for both analog and digital signaling.

Guided Media: Fiber Optics

Medium consists of **hair-width strand of silicon or glass**. Signal is pulses of light. Pulse = "1", no pulse = "0".

Three Concentric Sections:

CORE - Innermost, solid dielectric cylinder, refractive index n_1

CLADDING - Refractive index $n_2 < n_1$, enables total internal reflection

JACKET - Plastic, protects against moisture, abrasion, crushing

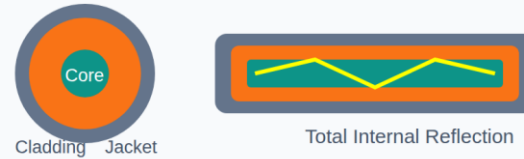


Figure: Optical Fiber

Three Components Required:

1. **Fiber Medium** - carries light for 100s of km
2. **Light Source** - LED or laser diode
3. **Light Detector** - Photo diode

Fiber Optics: Advantages

1. HIGH DATA RATE

1000 Mbps (1 Gbps) over kilometers. Error rates almost negligible.

2. SECURITY

Difficult to tap, hard for unauthorized access. Higher reliability.

3. THIN & LIGHTWEIGHT

Much thinner than copper circuits. Easier to install.

4. NO INTERFERENCE

Not susceptible to electrical interference or corrosion.

5. GREATER DISTANCE

Greater repeater distance than coax (avg 8 miles).

FIBER USES:

Long-haul trunks • Metropolitan trunks • Rural exchange • Local loops • High-speed LAN (100Mbps)

Disadvantage: Point-to-point, difficult to tap, special training required, two fibers for full duplex.

Unguided Transmission (Wireless)

Used when running physical cable between two points is not possible. Example: buildings separated by public street.

Difficulties:

1. Weather interferes with signals (clouds, rain, lightning)
2. Radio transmissions easy to tap (security concern)
3. Signals bouncing off structures cause out-of-phase signals

TYPES OF WIRELESS TRANSMISSION

Radio Waves

Omnidirectional, can penetrate walls

Microwaves

Directional, line-of-sight transmission

Infrared

Short range, cannot penetrate walls

Noise in Communication

As signal is transmitted, undesired signal (**noise**) gets mixed with it. Noise has four types:

1. THERMAL NOISE

Thermal agitation of electrons. Called **white noise**.

2. INTERMODULATION

Signals f_1 and f_2 generate (f_1+f_2) and (f_1-f_2) .

3. CROSS TALK

EM radiation induced on conductors due to proximity.

4. IMPULSE NOISE

Lightning, loose contacts. Primary source of **bit-errors**.

Effects: Noise limits bandwidth and data rate.

Nyquist Bit Rate (Noiseless Channel)

Maximum rate at which data can be communicated is known as **channel capacity**.

For **noise-free channel** with Bandwidth B , maximum data rate is $2B$ (due to intersymbol interference).

If signal has m discrete levels:

$$C = 2B \log_2 m \text{ bits/sec}$$

C = Channel capacity, B = Bandwidth, m = Signal levels

Example 1.1

Telephone channel $B = 4$ kHz. No noise. Find capacity for: (a) 2 levels (b) 128 levels

Solution:

(a) $m = 2$:

$$C = 2B = 2 \times 4000 = 8 \text{ Kbits/s}$$

(b) $m = 128$:

$$C = 2 \times 4000 \times \log_2 128 = 8000 \times 7 = 56 \text{ Kbits/s}$$

Baud Rate and Bit Rate

BAUD RATE

Number of **distinct symbols** transmitted per second, irrespective of encoding.

One signal unit may represent one or more bits.

$$\begin{aligned}\text{Max Baud Rate} &= 1/\text{Element Width} \\ &= 2B\end{aligned}$$

Key Relationship:

For **binary encoding**, bit rate = baud rate
(I = Baud Rate)

BIT RATE (Information Rate)

Actual equivalent number of **bits transmitted per second**.

$$I = \text{Baud Rate} \times \text{Bits per Baud}$$

$$I = \text{Baud Rate} \times \log_2 m$$

Example:

Baud rate = 1000, m = 4 levels:

$$\begin{aligned}I &= 1000 \times \log_2 4 = 1000 \times 2 = \\ &2000 \text{ bps}\end{aligned}$$

What is Baud Rate?

Baud rate is the **number of signal units (symbols) transmitted per second**.
One signal unit may represent **one or more bits**.

Unit

- baud (symbols per second)

Baud rate shows **how many times the signal changes per second**.

Relationship Between Bit Rate and Baud Rate

$$\text{Bit Rate} = \text{Baud Rate} \times \text{Bits per Symbol}$$

Where:

$$\text{Bits per Symbol} = \log_2 L$$

L = number of signal levels

Case 1: When Bit Rate = Baud Rate

If:

- Each signal carries **1 bit**
- Binary signaling ($L = 2$)

Then:

$$\text{Bit Rate} = \text{Baud Rate}$$

Example

- 1000 baud
- 1 bit per symbol

Therefore, Bit rate = **1000 bps**

Case 2: When Bit Rate $\neq$ Baud Rate

If:

- One signal carries **multiple bits**
- Multi-level signaling

Example 1

Baud rate = 1000 baud

Signal levels = 4 ($L = 4$)

$$\text{Bits per symbol} = \log_2 4 = 2$$

$$\text{Bit Rate} = 1000 \times 2 = 2000 \text{ bps}$$

Example 2 (Modern Systems)

- Baud rate = 2400 baud
- Bits per symbol = 4

Therefore, Bit rate = **9600 bps**

Why Baud Rate Is Lower in Modern Systems

- ✓ High baud rate → more bandwidth needed
- ✓ High baud rate → more noise & distortion

So modern systems:

- Keep **baud rate** low
- Increase **bits per symbol**

Generally,

Bit Rate $\geq$ Baud Rate
(Bit rate equals baud rate only for binary signaling.)

Shannon Capacity (Noisy Channel)

In presence of Gaussian white noise,
Shannon-Hartley theorem gives maximum data rate:

$$C = B \log_2 (1 + S/N)$$

C = Capacity (bps), **B** = Bandwidth (Hz), **S/N** = Signal-to-Noise ratio

S/N in dB: $10 \times \log_{10}(S/N)$

NOTE: Between Nyquist and Shannon, the smallest value establishes the limit.

Example 1.2: B = 4 KHz, S/N: (a) 20 dB (b) 30 dB (c) 40 dB

(a) $S/N=100 \rightarrow C = 4 \times 10^3 \times \log_2(101) = 26.6 \text{ kbits/s}$

(b) $S/N=1000 \rightarrow C = 39.8 \text{ kbits/s}$

(c) $S/N=10000 \rightarrow C = 53.1 \text{ kbits/s}$

Example 1.3: B = 4 KHz, S/N = 30 dB, 4-level encoding

Nyquist = $2 \times 4000 \times \log_2 4 = 16 \text{ Kbps}$

Shannon = 39.8 Kbps

Information capacity I = 16 Kbps (smaller)

Solved Examples

Example 1.4: $B = 4$ kHz, $S/N = 30$ dB, 128-level

Nyquist: 56 kbits/s | Shannon: 39.8 Kbps $\rightarrow$ **C = 39.8 Kbps**

Example 1.5: 160 kbps at $B = 20$ kHz

(a) $M = 16$ levels | (b) $S/N = 24.07$ dB

Example 1.6: $B = 4$ kHz, no noise, 4 levels

$$I = 2 \times 4000 \times \log_2 4 = 16 \text{ Kbps}$$

Example 1.7: $B = 10$ MHz, $S/N = 60$ dB

$$C = 10 \times \log_2(1 + 10^6)$$

Summary

DATA COMMUNICATION

Source → Transmitter → Medium → Receiver → Destination

SIGNALS

Analog (continuous) vs Digital (0,1). Bandwidth $B = F_h - F_l$

TRANSMISSION MEDIA

Guided: Twisted Pair, Coaxial, Fiber

Unguided: Radio, Microwave, Infrared

NOISE TYPES

Thermal, Intermodulation, Cross-talk, Impulse

KEY FORMULAS

Nyquist: $C = 2B \log_2 m$

Shannon: $C = B \log_2(1 + S/N)$

Bit Rate: $I = \text{Baud Rate} \times \log_2 m$

BASEBAND vs BROADBAND

Baseband: Digital, bi-directional

Broadband: Analog, FDM, unidirectional

FIBER ADVANTAGES

High data rate (1 Gbps), security, no interference